



Induction of *decision rules*

LEM2 (***L***earning from ***E***xamples ***M***odule, version ***2***)



Algorithms for induction of decision rules

- *GTS*
- *LEM1* / *LEM2*
- *CN2*
- *AQ*
- *INLEM*
- *MODLEM*
- ...



LEM2 algorithm

No.	Inflation	Budget_deficit	Currency_reserves	Interest_rates
1	decrease	no_change	increase	reduction
2	decrease	increase	decrease	reduction
3	no_change	increase	increase	increase
4	no_change	increase	decrease	reduction
5	no_change	no_change	decrease	reduction
6	no_change	increase	no_change	increase
7	decrease	increase	increase	reduction
8	no_change	no_change	increase	increase

We determine the set ***B*** of all cases of the **reduction** class

Thus ***B*** = {**1**, **2**, **4**, **5**, **7**}

No.	Inflation	Budget_deficit	Currency_reserves	Interest_rates
1	decrease	no_change	increase	reduction
2	decrease	increase	decrease	reduction
3	no_change	increase	increase	increase
4	no_change	increase	decrease	reduction
5	no_change	no_change	decrease	reduction
6	no_change	increase	no_change	increase
7	decrease	increase	increase	reduction
8	no_change	no_change	increase	increase

We determine the set $T(G)$ of all possible attribute-value pairs from the cases of the set B

Thus $T(G) = \{(Inflation, decrease), (Inflation, no_change), (Budget_deficit, no_change), (Budget_deficit, increase), (Currency_reserves, increase), (Currency_reserves, decrease)\}$

1	decrease	no_change	increase	reduction
2	decrease	increase	decrease	reduction
3	no_change	increase	increase	increase
4	no_change	increase	decrease	reduction
5	no_change	no_change	decrease	reduction
6	no_change	increase	no_change	increase
7	decrease	increase	increase	reduction
8	no_change	no_change	increase	increase

We're preliminary assuming that:

$$B = G = \{1, 2, 4, 5, 7\} \text{ and } \pi = \emptyset$$

Furthermore:

$$[(\text{Inflation}, \text{decrease})] = \{1, 2, 7\}$$

$$[(\text{Inflation}, \text{no_change})] = \{3, 4, 5, 6, 8\}$$

$$[(\text{Budget_deficit}, \text{no_change})] = \{1, 5, 8\}$$

$$[(\text{Budget_deficit}, \text{increase})] = \{2, 3, 4, 6, 7\}$$

Thus $T(G) = \{(\text{Inflation}, \text{no_change}), (\text{Budget_deficit}, \text{no_change}), (\text{Budget_deficit}, \text{increase}), (\text{Currency_reserves}, \text{increase}), (\text{Currency_reserves}, \text{decrease})\}$

Since the following condition is fulfilled

$$[T] = [(\text{Inflation}, \text{decrease})] = \{1, 2, 7\} \subset B$$

The decision rule **R1** can be created (induced)



Rule 1:

IF Inflation **IS** decrease

THEN Interest_rates **IS** reduction

$$\pi = \pi \cup \{T\} = \emptyset \cup \{1, 2, 7\} = \{1, 2, 7\}$$

$$G = B - [\pi] = \{1, 2, 4, 5, 7\} - \{1, 2, 7\} = \{4, 5\}$$

Furthermore:

$$[(Inflation, no_change)] = \{3, 4, 5, 6, 8\}$$

$$[(Budget_deficit, no_change)] = \{1, 5, 8\}$$

$$[(Budget_deficit, increase)] = \{2, 3, 4, 6, 7\}$$

$$[(Currency_reserves, increase)] = \{1, 3, 7, 8\}$$

$$[(Currency_reserves, decrease)] = \{2, 4, 5\}$$

Thus $T(G) = \{(Inflation, no_change), (Budget_deficit, no_change), (Budget_deficit, increase), (Currency_reserves, increase)\}$

Since the following condition is fulfilled

$$[T] = [(Currency_reserves, decrease)] = \{2, 4, 5\} \subset B$$

The new decision rule **R2** can be induced



Rule 1:

IF Inflation **IS** decrease

THEN Interest_rates **IS** **reduction**

Rule 2:

IF Currency_reserves **IS** decrease

THEN Interest_rates **IS** **reduction**

$$\pi = \pi \cup \{T\} = \{1, 2, 7\} \cup \{4, 5\} = \{1, 2, 4, 5, 7\}$$

$$G = B - [\pi] = \{1, 2, 4, 5, 7\} - \{1, 2, 4, 5, 7\} = \emptyset$$

Since **G** = **∅**, therefore, the above decision rules cover all cases from the **reduction** category. Thus, now the decision rules covering the next category (i.e., **increase**) need to be found

We determine the set ***B*** of all cases of the ***increase*** class

Therefore, ***B*** = {***3, 6, 8***}

No.	Inflation	Budget_deficit	Currency_reserves	Interest_rates
1	decrease	no_change	increase	reduction
2	decrease	increase	decrease	reduction
3	no_change	increase	increase	increase
4	no_change	increase	decrease	reduction
5	no_change	no_change	decrease	reduction
6	no_change	increase	no_change	increase
7	decrease	increase	increase	reduction
8	no_change	no_change	increase	increase

We determine the set $T(G)$ of all possible attribute-value pairs from the cases of the set B

Thus $T(G) = \{ (Inflation, no_change), (Budget_deficit, no_change), (Budget_deficit, increase), (Currency_reserves, increase), (Currency_reserves, no_change) \}$

1	decrease	no_change	increase	reduction
2	decrease	increase	decrease	reduction
3	no_change	increase	increase	increase
4	no_change	increase	decrease	reduction
5	no_change	no_change	decrease	reduction
6	no_change	increase	no_change	increase
7	decrease	increase	increase	reduction
8	no_change	no_change	increase	increase

Furthermore:

$[(Inflation, no_change)] = \{3, 4, 5, 6, 8\}$

$[(Budget_deficit, no_change)] = \{1, 5, 8\}$

$[(Budget_deficit, increase)] = \{2, 3, 4, 6, 7\}$

$[(Currency_reserves, increase)] = \{1, 3, 7, 8\}$

$[(Currency_reserves, no_change)] = \{6\}$

From all pairs we choose those which cover the largest number of cases from the *increase* class

$[T] = [(Inflation, no_change)]$ covers 3 cases

6	no_change	increase	no_change	increase
7	decrease	increase	increase	reduction
8	no_change	no_change	increase	increase

Since condition $[(Inflation, no_change)]$ covers
a.) cases 3, 6, 8 from the *increase* class, and
b.) cases 4, 5 from the *reduction* class at the same
time, i.e.:

$$[T] = [(Inflation, no_change)] \not\subset B$$

Thus in the next iteration an additional condition
should be found to eliminate cases 4, 5.

From the set $T(G)$ we remove the condition
 $[(Inflation, no_change)]$

Therefore: $T(G) = \{(Budget_deficit, no_change),$
 $(Budget_deficit, increase), (Currency_reserves,$
 $increase), (Currency_reserves, no_change)\}$



LEM2 algorithm

Therefore:

$[(Budget_deficit, no_change)] = \{1, 5, 8\}$

$[(Budget_deficit, increase)] = \{2, 3, 4, 6, 7\}$

$[(Currency_reserves, increase)] = \{1, 3, 7, 8\}$

$[(Currency_reserves, no_change)] = \{6\}$

Thus: $T(G) = \{(Budget_deficit, no_change), (Budget_deficit, increase), (Currency_reserves, no_change)\}$ and
 $T = \{(Inflation, no_change), (Currency_reserves, increase)\}$

Outcome: $[T] = \{3, 8\}$



Rule 1:

IF Inflation **IS** decrease

THEN Interest_rates **IS** reduction

Rule 2:

IF Currency_reserves **IS** decrease

THEN Interest_rates **IS** reduction

Rule 3:

IF Inflation **IS** no_change

AND Currency_reserves **IS** increase

THEN Interest_rates **IS** increase

$$\pi = \pi \cup \{T\} = \emptyset \cup \{3, 8\} = \{3, 8\}$$

$$G = B - [\pi] = \{3, 6, 8\} - \{3, 8\} = \{6\}$$



LEM2 algorithm

Thus:

$[(Budget_deficit, no_change)] = \{1, 5, 8\}$

$[(Budget_deficit, increase)] = \{2, 3, 4, 6, 7\}$

$[(Currency_reserves, no_change)] = \{6\}$

Thus: $T(G) = \{(Budget_deficit, no_change), (Budget_deficit, increase)\}$

and

$[T] = [(Currency_reserves, no_change)] \subset B$ – thus,
the decision rule **R4** can be created

Rule 1:

IF Inflation IS decrease

THEN Interest_rates IS reduction

Rule 2:

IF Currency_reserves IS decrease

THEN Interest_rates IS reduction

Rule 3:

IF Inflation IS no_change

AND Currency_reserves IS increase

THEN Interest_rates IS increase

Rule 4:

IF Currency_reserves IS no_change

THEN Interest_rates IS increase



$$\pi = \pi \cup \{T\} = \{3, 8\} \cup \{6\} = \{3, 6, 8\}$$

$$G = B - [\pi] = \{3, 6, 8\} - \{3, 6, 8\} = \emptyset$$

$G = \emptyset$ – thus, all cases from the *increase* category were covered by the decision rules

The algorithm ends



*Thank you for your
attention...*

